

Dhanush Sambasivam

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Education

University of Maryland

Master's in Data Science (GPA: 3.86/4)

Coursework: Database Systems, Data Science, Machine Learning, Statistical Modeling, AI, Data Visualization, Applied Probability

Aug 2024 – Current

College Park, MD

Vellore Institute of Technology

Bachelor of Technology in Computer Science (GPA: 3.86/4)

Jun 2020 – May 2024

Chennai, India

Technical Skills

- **Languages:** Python, R, MATLAB, SAS, C++, Java
- **Libraries:** Matplotlib, NumPy, Pandas, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch, MLflow, NLTK, Hugging Face
- **Technologies:** Marketing Analytics, Snowflake, Databricks, Healthcare Analytics, Power BI, Tableau, GCP, Spark, AWS, A/B Testing, CI/CD Pipelines, Git
- **Certifications:** Generative AI with LLM (Deeplearning.AI), Data Science Professional Certificate (IBM), PowerBI Virtual Experience (PwC Forage), Certified in Python, C, C++ (IIT, Bombay), Finance Business Analyst (Finlatics)

Work Experience

University of Maryland

Data Scientist, Counseling and Student Services

Sep 2024 – Present

College Park, Maryland

- Analyzed student appointments, counseling engagement, and retention data across **5 departments** using Python and SQL, identifying behavioral trends and metric inconsistencies that improved reporting logic and accuracy by **40%**.
- Built Snowflake SQL pipelines using CTEs, materialized views, and query profiling to transform and aggregate **50K+** student engagement records, reducing Power BI dashboard refresh time by **35%**.
- Automated end-to-end institutional reporting workflows by integrating Python and Power Automate across **5 intake** systems, reducing weekly reporting turnaround from **10 hours to under 3 hours**.
- Architected an end-to-end NLP pipeline using fine-tuned BERT for sentiment classification and LDA for topic modeling on **5,000+** anonymized counseling records, reducing manual counselor review time by **70%**.

Amgen

Data Scientist, Media Investment Team, Commercial Data & Analytics

May 2025 – Sept 2025

Thousand Oaks, California

- Engineered scalable ETL pipelines using PySpark, SQL, and BigQuery to integrate **15+ media ad** sources across **10M+ records**, enabling Marketing Mix Modeling while reducing pipeline processing time by **8x** across the U.S. brand portfolio.
- Created reusable transformation workflows and SQL views in Databricks to standardize leads, sales, and media KPIs, reducing manual data preparation by **75%**.
- Developed a behavioral HCP segmentation model (K-Means) on engagement and prescribing patterns across U.S. territories, identifying underperforming media channels and improving marketing budget allocation efficiency by **25%**.
- Engineered a Customer Lifetime Value forecasting framework using LightGBM with HCP prescription behavior features to score and prioritize retention targets, improving next-best-action targeting precision by **30%**.
- Consolidated **6+** Tableau dashboards into a unified cross-channel analytics dashboard, supporting marketing investment decisions and decreasing reporting errors by **40%**.

Neurealm (Previously GAVS Technologies)

Data Scientist, Intelligent Automation and Analytics Team

Aug 2023 – Jul 2024

Chennai, India

- Implemented a GenAI image generation pipeline using Stable Diffusion, Python, and LangChain, building a web interface for real-time image inference that reduced design turnaround time by **30%**.
- Built an enterprise QA system using LangChain RAG pipelines, OpenAI APIs, and FAISS vector embeddings over internal knowledge bases, improving retrieval accuracy by **25%** and reducing manual document search across **3 business units**.
- Designed a Python automation tool using Jira APIs to extract and transform operational ticket data into structured reporting pipelines, eliminating **8+ hours** of manual bi-weekly processing.
- Assembled Power BI dashboards integrating ServiceNow ticket data using Python and SQL to track SLA compliance, ticket trends, and resolution bottlenecks, reducing incident resolution time by **30%**.

National University of Singapore

Research Intern

May 2022 – Jul 2022

Singapore, Singapore

- Led a team of **5** in an NUS × Hewlett Packard Enterprises research initiative, performing feature engineering and identifying **20 high-signal** macroeconomic indicators (trade freedom, fiscal policy, monetary stability) for model performance.
- Constructed a time-series analytics framework using Python and Tableau to analyze Economic Freedom Index trends across **180 countries** over **10 years**, enabling econometric analysis across income groups and regions.
- Evaluated **10+ machine learning** models using stratified cross-validation and ROC-AUC scoring, achieving 93% predictive accuracy with XGBoost and deploying the forecasting pipeline as a Streamlit application on AWS.

Projects

AI Football Analytics using Multi-Agent YOLO System (NVIDIA Hackathon)

Oct 2025

- Developed an automated football analytics system using YOLOv8 for real-time player tracking, integrated with a multi-agent MCP framework using LangChain to analyze player movement with **35%** faster processing.

Money Laundering Detection through Financial Data Analysis

Sep 2024

- Implemented scalable ETL pipelines with PySpark and SQL to process 190M+ financial transactions across 10 years; applied anomaly detection models (Isolation Forest, XGBoost) for money laundering risk scoring with 95% detection accuracy on GCP.